JAIN DEEMED-TO-BE UNIVERSITY

आविष्कार FET HACKATHON

FET Alumni and Outreach — CONNECT INNOVATE IMPACT

SIGNFORGE

From a few native signs to a model-ready corpus

Low-resource sign-language corpus builder · Round 2: the working build

10 app pages, live	**33** signs · ISL + ASL	**78%** ISL, held-out	**83** automated tests

Team Claude's Clan — Tashvi Adhlakha (Team Leader), Bhavesh Lakhmani, Reuben Raju, Aakarshi Dwivedi

Live demo bhaveshlakhmani25-ui.github.io/signforge

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01 · THE PROBLEM

Problem Statement

> *Low-resource sign-language corpus builder using motion-capture transfer learning*
>
> *Official problem statement · Aavishkar FET Hackathon*

WHY IT IS HARD TODAY

Scarce, scattered data

120 datasets across 35 sign languages, in different formats and licences.

Labels don't line up

No shared standard for glosses, splits or quality, so results aren't comparable.

Too few signers

One or two signers per sign; models break on new people and regional variants.

Consent is an afterthought

Raw video of real people kept without clear permission or withdrawal.

70M+	Deaf people worldwide, using 300+ sign languages (UN / WFD)
ISL	Has datasets (INCLUDE, iSign), but they are small, uneven and hard to extend
?	What data should we collect next? No tool answers this today.

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02 · WHAT WE BUILT

Solution



WHAT IT PRODUCES

A model-ready corpus
Landmarks + manifest, with consent, licence and review status per clip.

Models you can trust
Frozen test set, macro-F1 and calibration; 'not sure' when unsure.

A plan for what to record
Signs ranked by coverage, signer and model gaps, each with a reason.

8 working pages: Capture · Corpus · Queue · Models · Interpret · Learn · Export · Dashboard

03 · HOW IT WORKS

Tech Stack + Workflow

Next.js | React | TypeScript | Tailwind | MediaPipe | Python | FastAPI | PyTorch | NumPy | SQLite | GH Actions

IN THE BROWSER · VIDEO STAYS ON DEVICE

Camera / video → MediaPipe Holistic → 75 landmarks

BACKEND · FASTAPI + PYTORCH

QC + duplicates → SQLite + .npz → GRU train / eval → Queue + review

MODEL PIPELINE · IDENTICAL FOR TRAINING, LIVE CAMERA AND THE IN-BROWSER PORT

Frames	Landmarks	27 points	30 frames	Normalise	GRU 128	Calibrate
camera	33 + 2×21	x, y	10 fps	shoulders	dropout 0.3	temperature

Privacy by pose: only landmarks leave the browser. The hosted demo runs the same model and QC in TypeScript, checked against Python.

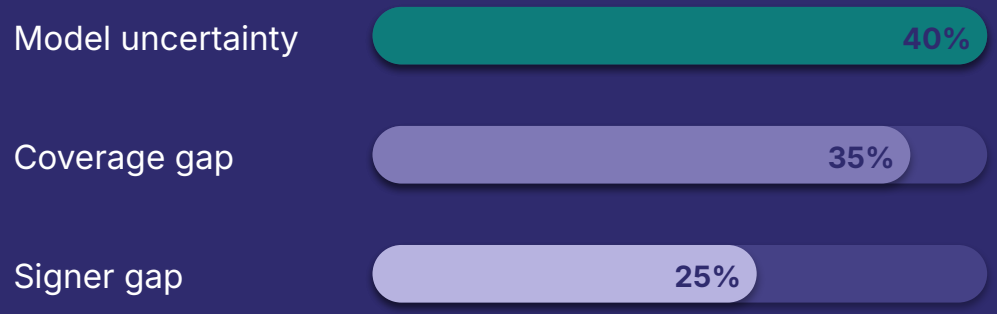
04 · WHAT IS NEW

Innovation

Model-guided collection

our core innovation

Instead of recording more of everything, SIGNFORGE ranks every sign by how much a new clip would help, and says why.



Output: “Record ‹sign› from a new signer: only 1 signer represented, model unsure.” One click opens Capture.

A model check that admits doubt

Same-language model, calibrated scores, 'not sure' below 55%, no auto-accept until reliable.

Privacy by pose, real consent

Landmarks only; withdrawing consent deletes clips and models trained on them.

Runs without a server

Model, quality check and retraining run in the browser, verified clip for clip.

Teach it new signs

Add a sign, record 3+ approved clips, retrain in seconds; Interpret knows it.

Learners grow the corpus

Practise against real dataset clips; opted-in attempts go to review.

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05 · WHAT IT CHANGES

Impact

Better data
consented, checked, reviewed

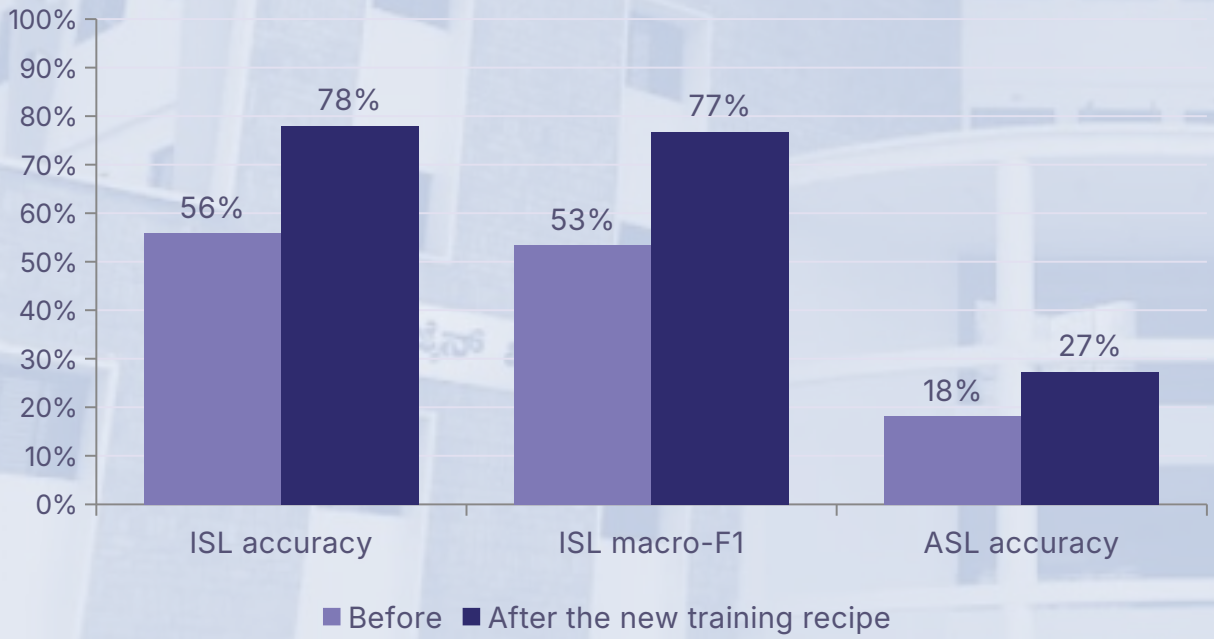
→

Better models
measured on frozen test sets

→

Accessible apps
built on documented corpora

Measured on the same frozen test clips



56% → 78%
ISL accuracy on the same 59 held-out clips

0.11 → 0.07
calibration error, now passes the 0.10 check

22 clips
from 8 consenting team signers

Who benefits: Deaf communities steer what is collected; researchers get reusable data; learners practise free. Downstream impact is not yet measured.

06 · WHO IT IS FOR

Target Audience / Market

WHO	THEIR PROBLEM	WHAT SIGNFORGE GIVES THEM	PAGES THEY USE
Research labs	Small, inconsistent datasets; results can't be compared	A documented corpus and a frozen-split benchmark, exportable	Models · Export
Accessibility startups	Need the exact signs their product uses	A ranked collection queue, with consent tracked per clip	Capture · Queue
Deaf organisations	Data taken from the community without a say	Run collection and review; consent and withdrawal built in	Review · Consent
Public institutions	ISL access in hospitals, transport, offices	Domain vocabularies built and validated with signers	Capture · Corpus
AI companies	Need licensable, provenance-clean data	Licence and provenance per clip; landmarks, not raw video	Export

Beachhead: Indian Sign Language with INCLUDE as the open benchmark. ASL already runs, so any under-resourced sign language can be added as a project.

07 · HOW IT SUSTAINS ITSELF

Revenue Model

Community & academic

Free

Open-source, self-hosted

Research

Support plan

Setup, data audits, benchmarks

Pro

Subscription

Hosted collection, review, model runs

Enterprise

Licence

Private deployment, governance, SSO

Data services

Per project

Consented corpus building; contributors paid

more hosting, support and services →

Open core, paid hosting and services. Contributors are credited and paid; their data is never free labour. Billing is not part of the prototype.

08 · WHAT COMES NEXT

Future Improvements

	Now (built)	Next	Later
More data	✔ 835 clips: 17 ISL + 16 ASL signs, 22 team clips	Import INCLUDE People: 10 signs, archives checksummed; more signers per sign	Community collection at scale; regional variants
Better accuracy	✔ ISL 78% on held-out clips, calibrated	Verify a pretrained checkpoint for real transfer; target weak signs (ALRIGHT 0/4)	Signer-independent and continuous-signing evaluation
Platform & community	✔ Full loop, browser demo, new signs	Shared hosted backend; Deaf and native reviewer onboarding	Mobile capture, ELAN export, more sign languages

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Thank You

Questions?

Model uncertainty → next sample → better data

Try it bhaveshlakhmani25-ui.github.io/signforge

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